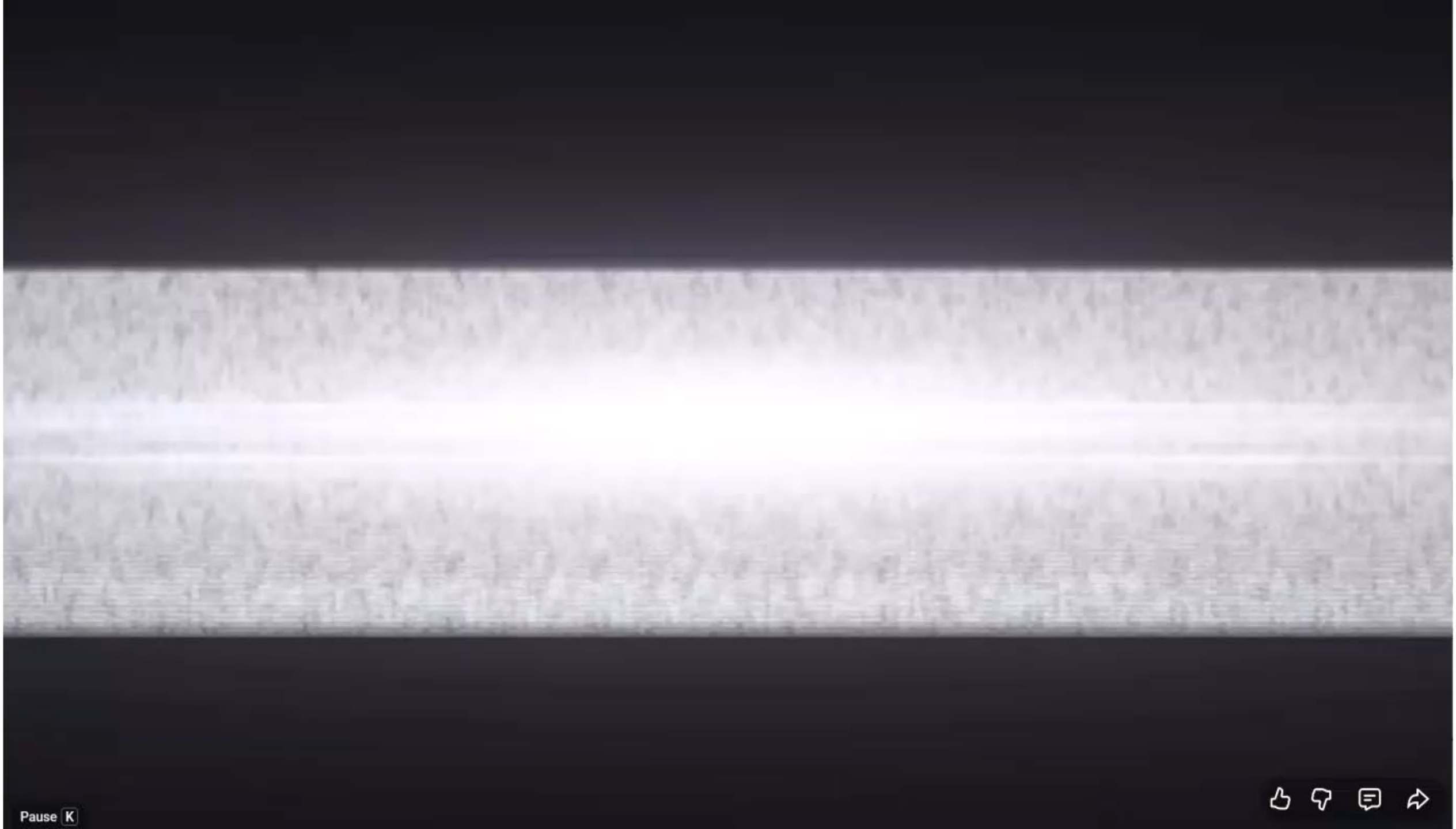
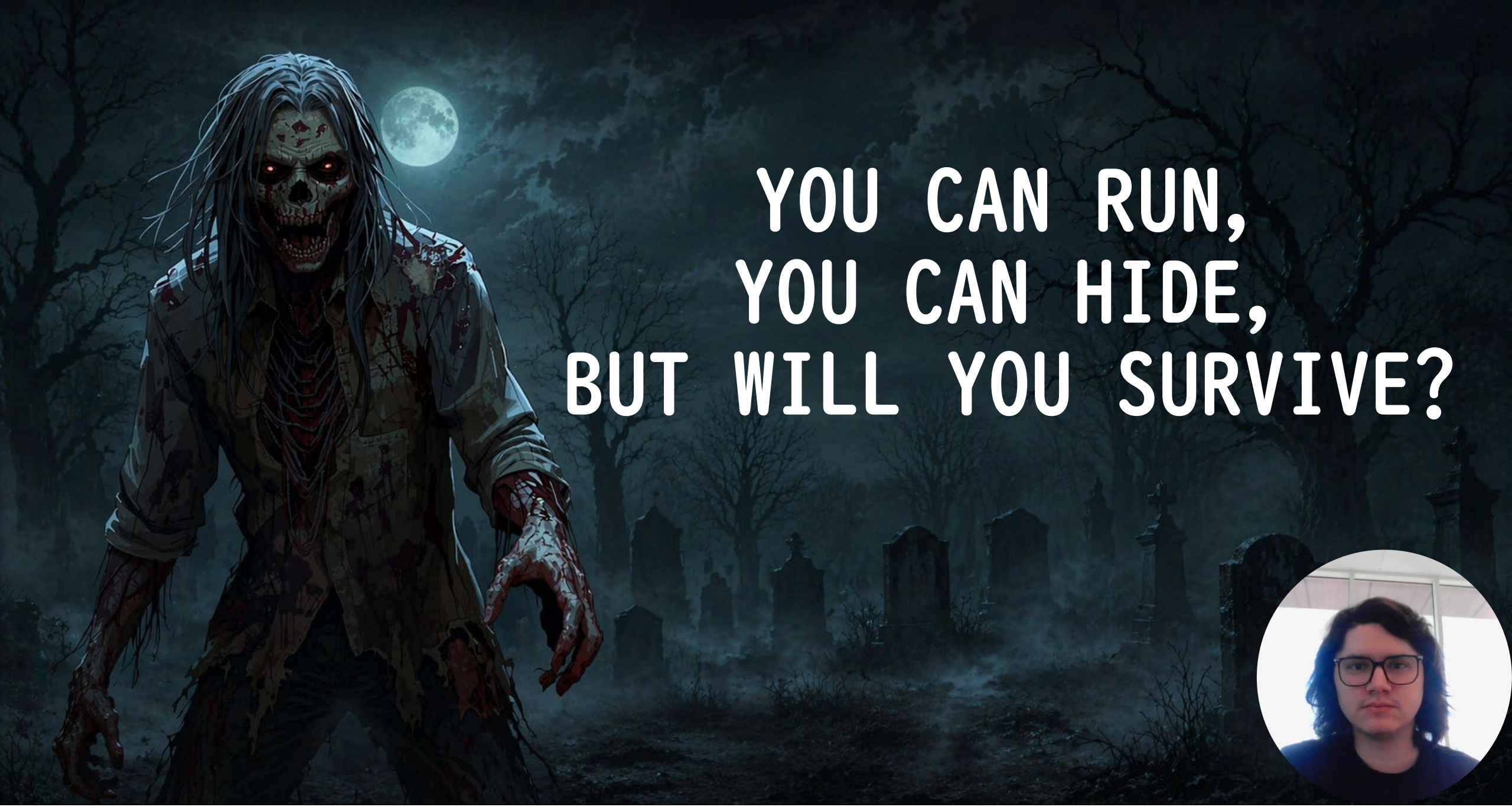

Pause K



YOU CAN RUN, YOU CAN HIDE, BUT WILL YOU SURVIVE?

Team Members: Curtis Moore | Kiana Whitney | Lennen Madera | Rebecca Rickard

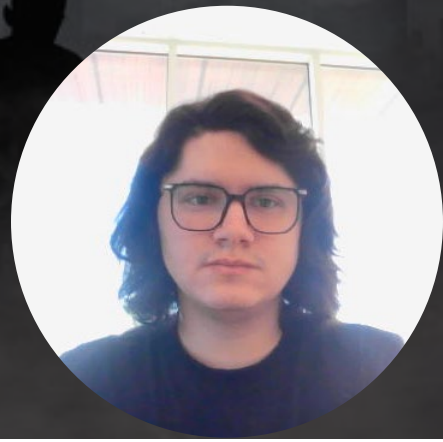
Spring 2025 – Big Data Analytics and Management - Final Project

WARNING

LIVE COVERAGE: FICTION HAS BECOME REALITY. THE NEWS YOU NEED TO KNOW

We are using an adapted epidemiological model (Alemi, 2015) that we have overlaid with additional data to account for human factors.

We will get you the analysis so you can run or hide to stay alive.



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Team Members: Curtis Moore | Kiana Whitney | Lennen Madera | Rebecca Rickard

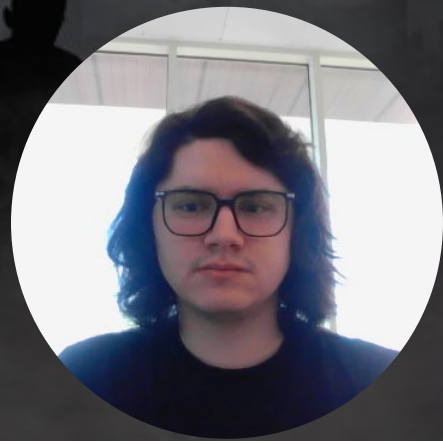
WHAT WE KNOW SO FAR

Focus: North Carolina

- Diverse geography creates varied conditions
- Mix of urban, suburban, and rural populations
- Temperate climate provides a balanced baseline for simulation
- Allows results to reflect a more general, real-world epidemiologic events

Questions Guiding Our Investigation

- What will the spread of the outbreak look like?
- How rapidly will it progress?
- Where are the most and least affected areas?
- How do human factors modify your odds of survival?



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UNDERSTANDING THE OUTBREAK

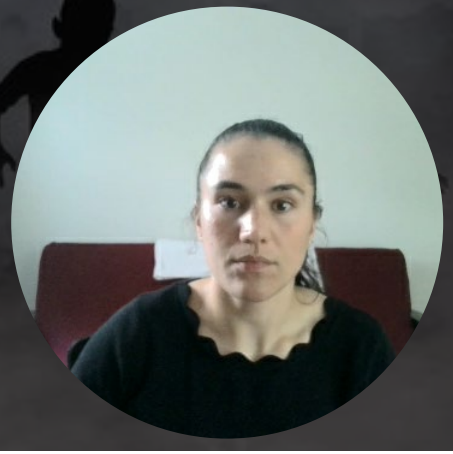
Fictional Parallel: The Last of Us

- Cordyceps Fungus
- Spread though bite
- 2-24 Incubation Period

Modeling the Spread

$$\dot{S} = -\beta SZ$$
$$\dot{Z} = (\beta - \kappa)SZ$$
$$\dot{R} = \kappa SZ$$

- S: Susceptible population/the uninfected humans
- Z: Infected or zombies
- R: Removed or terminated zombies
- $\beta = 6 \times 10^{-2}$: Bite parameter or rate at which an infected will bite a human
- $\kappa = 4 \times 10^{-2}$: Kill parameter which is the rate that a human kills a zombie



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THE HOPE FOR HUMANITY

Factors

- SC: Social and Community
- HP: Health and Physical Fitness
- IP: Infrastructure and Preparedness
- EG: Environmental and Geographic
- ME: Mobility and Escape
- EA: Education and Awareness

Human Survivability Index (HSI)

$$HSI = 0.25(SC) + 0.2(HP) + 0.15(IP) + 0.15(EG) + 0.15(ME) + 0.1(EA)$$

Modifying κ

$$\kappa_{\text{adjusted}} = \kappa * (1 + 0.5 * HSI)$$

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Social and Community

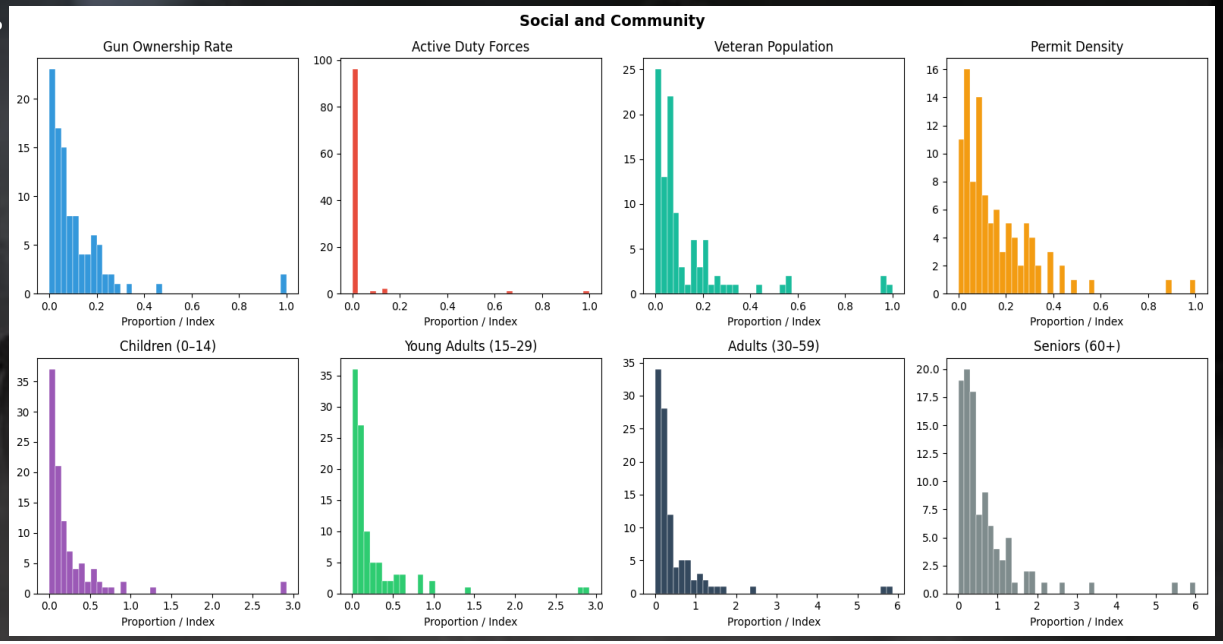
The composition and capabilities of local communities.

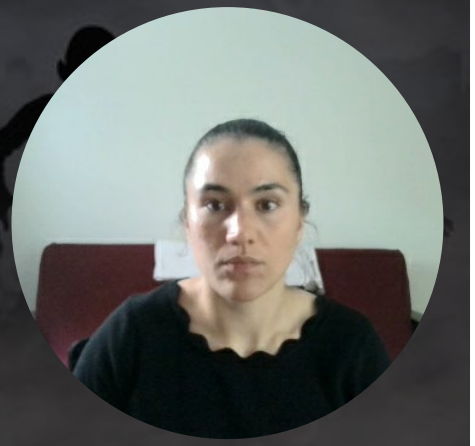
Datasets: U.S. Census (Age Groups (5–85+)), North Carolina Firearm Safety and Public Finance Review (Gun Ownership), North Carolina Wildlife Resources Commission (Hunting and Fishing License), Carolina Demography and North Carolina Department of Commerce (Military Personnel and Veteran Populations)

Assumptions:

- Used the NC Gun Safety Survey to align the country category of the 18+ population to estimate ownership
- Based on military personnel stations on major bases only

Limitations:

- Estimated gun ownership, as there are no state laws requiring the registration of firearms
- 



Health and Fitness

The physical capacity of the population to resist and respond to the outbreak.

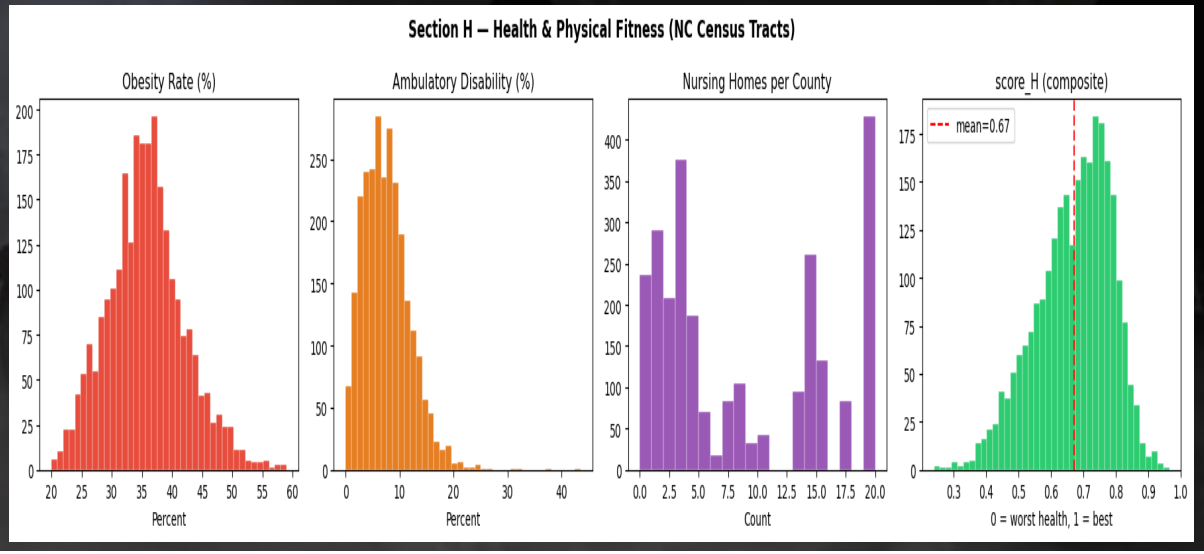
Datasets: CDC PLACES (Obesity Rate), ACS B18105 (Ambulatory Disability), CMS Care Compare (Nursing Home Providers)

Assumptions:

- Obesity rate used as tract-level proxy for overall physical fitness
- Ambulatory disability captures mobility-impaired populations at elevated zombie risk
- CMS nursing homes assigned to nearest county centroid via lat/lon; counts broadcast to all tracts

Limitations:

- All three factors are negative — higher impairment = lower score_H (inverted scoring)
- Nursing home data lacks GEOID; proximity-based county assignment may introduce minor spatial error



Infrastructure and Preparedness

The capacity of communities to prepare for and respond to the outbreak.

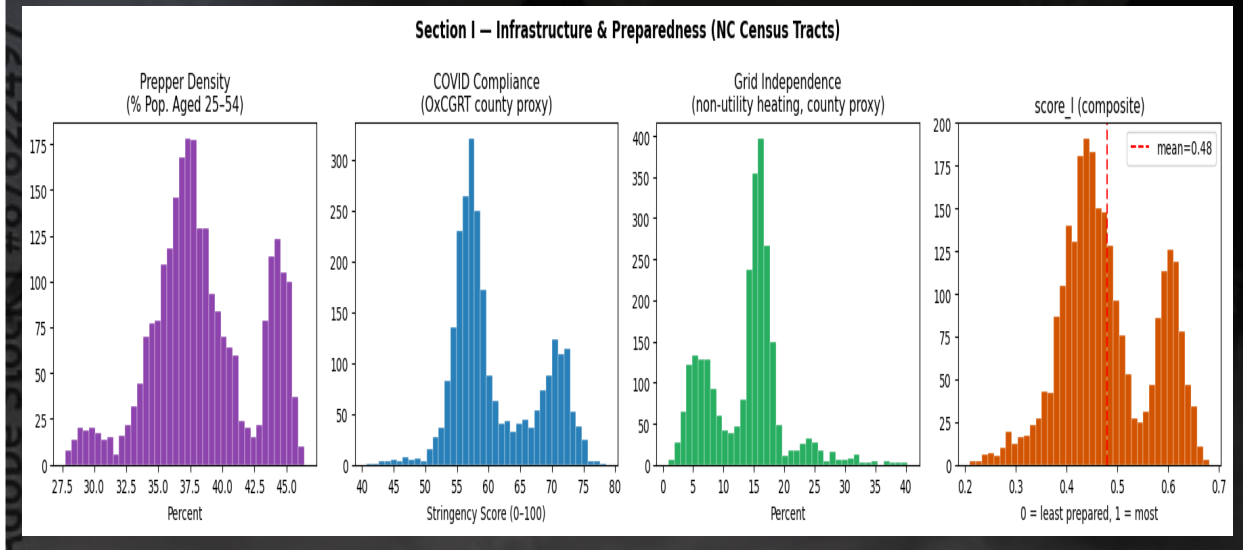
Datasets: ACS S0101 (% Population Aged 25–54), OxCGRT Stringency Index (Hale et al., 2021), ACS B25040 (House Heating Fuel)

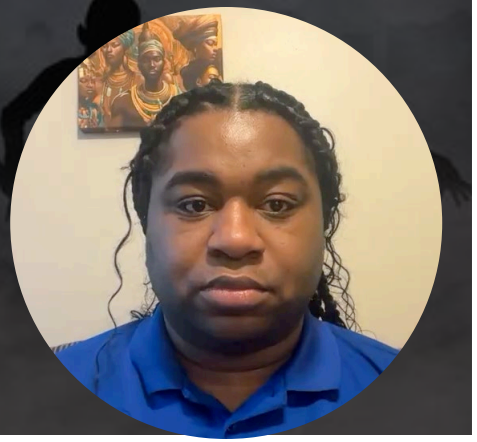
Assumptions:

- % population aged 25–54 used as proxy for prepper density (prime surviving-age cohort)
- COVID compliance estimated from OxCGRT national data mapped to NC metro vs. rural patterns
- Non-utility heating (wood, solar, bottled gas) used as proxy for grid independence

Limitations:

- OxCGRT does not publish sub-county data; county-level proxy applied to all tracts
- Grid independence estimated at county level from known NC rural/urban heating patterns





Environment and Geography

Latitude, longitude, elevation, and annual precipitation by county in NC

Dataset:

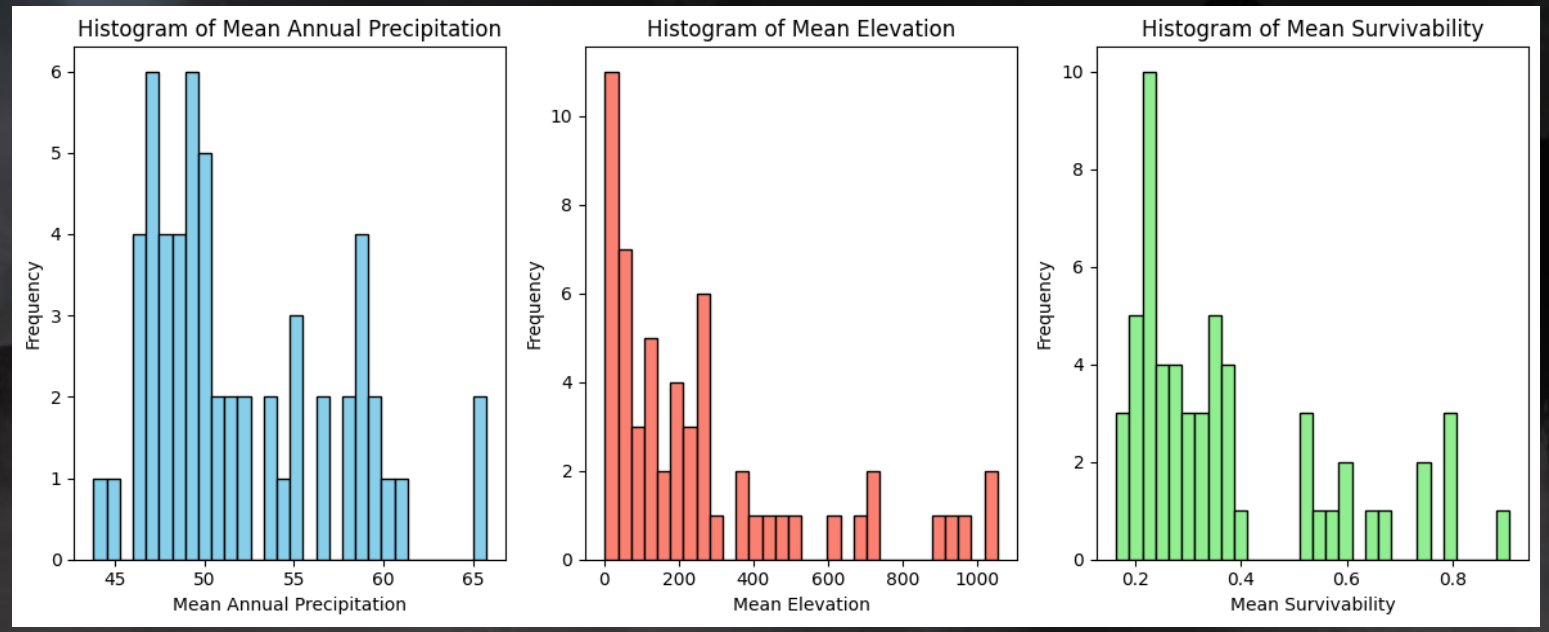
- NOAA Annual/Seasonal Data for NC Stations 1991 – 2010

Assumptions:

- Data collected between 1991 – 2010 should be similar to current seasonal conditions in NC due to the stability of NC's climate

Limitations:

- Number of weather stations per county were not consistent
- Most weather stations were newer and had a few years of data instead of the full 20 years



Mobility and Escape

The ability of populations to evacuate and reduce zombie encounters.

Datasets:

•ACS B25044 – Vehicle Availability by Tenure, USDA Rural-Urban Commuting Area (RUCA) Codes 2010

Assumptions:

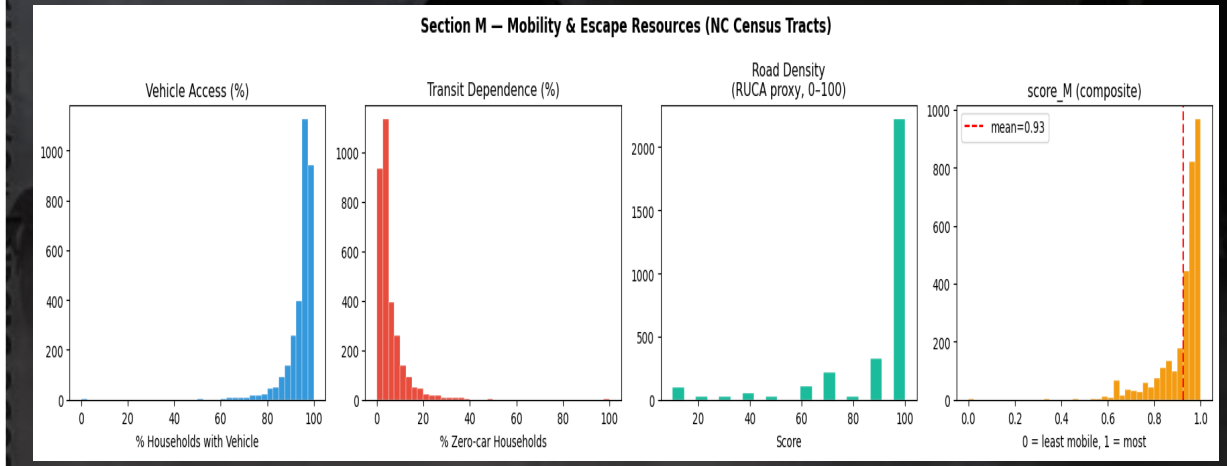
•% zero-vehicle households derived as transit dependence; inverted for vehicle access score

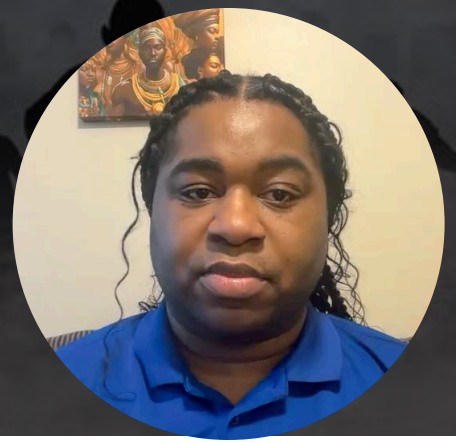
•RUCA codes (1–10) inverted to road density score: RUCA 1 (metro) = 100, RUCA 10 (rural) = 10

Limitations:

•RUCA code 99 (unpopulated tracts) excluded to prevent negative road density values

•Road density is an indirect proxy from RUCA classification, not actual road network data





Education and Awareness

The distribution of educational opportunities through NC.

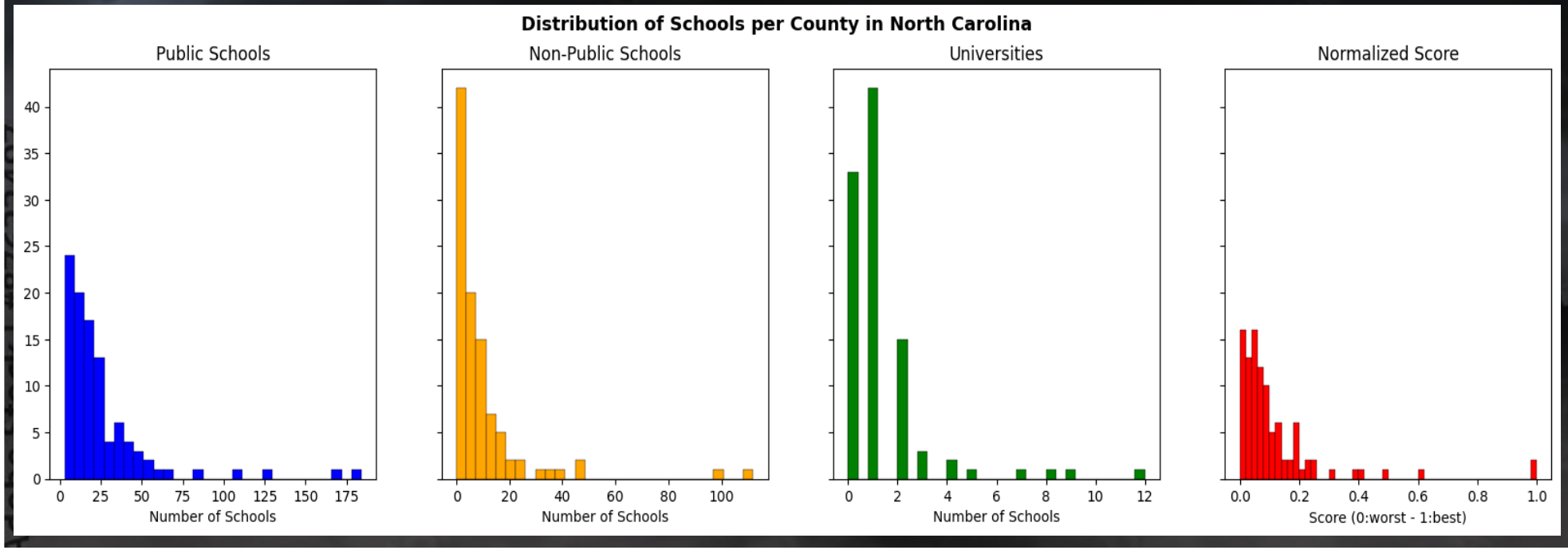
Datasets: Distribution of Public and Private Schools and Universities per each County

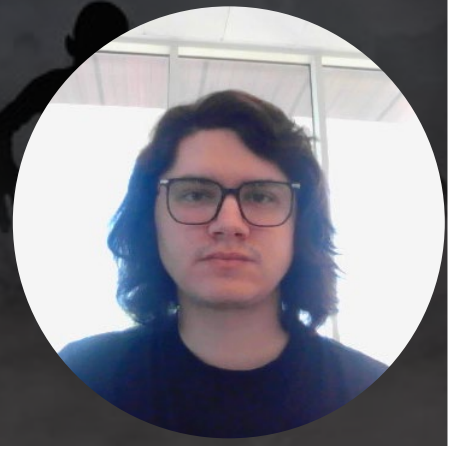
Assumptions:

- Educational availability is equivalent to its usage and the overall populations education
- A counties overall score is based on the density of education within that given county

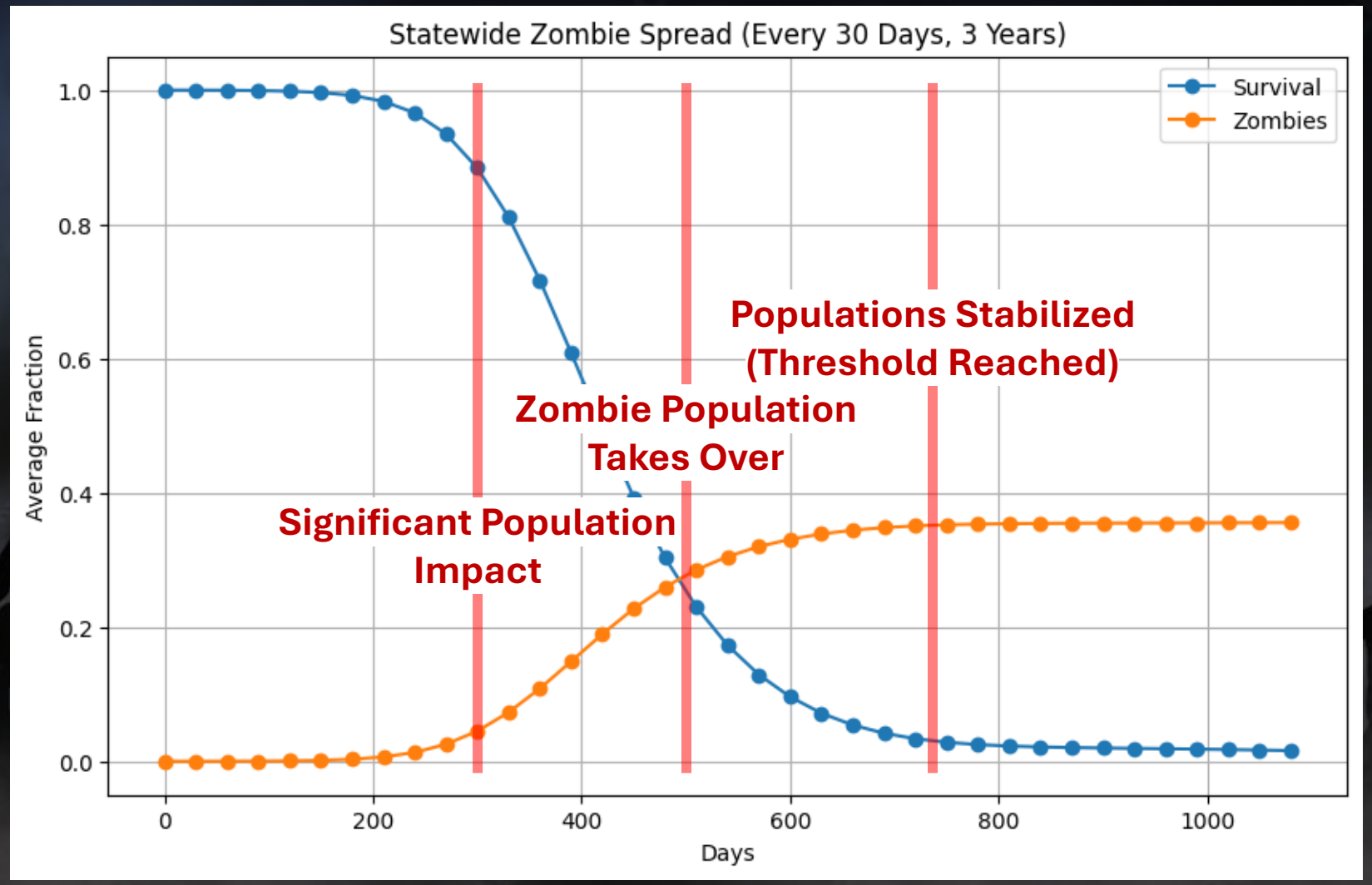
Limitations:

- Based on availability of education, rather than education received due to limited datasets available

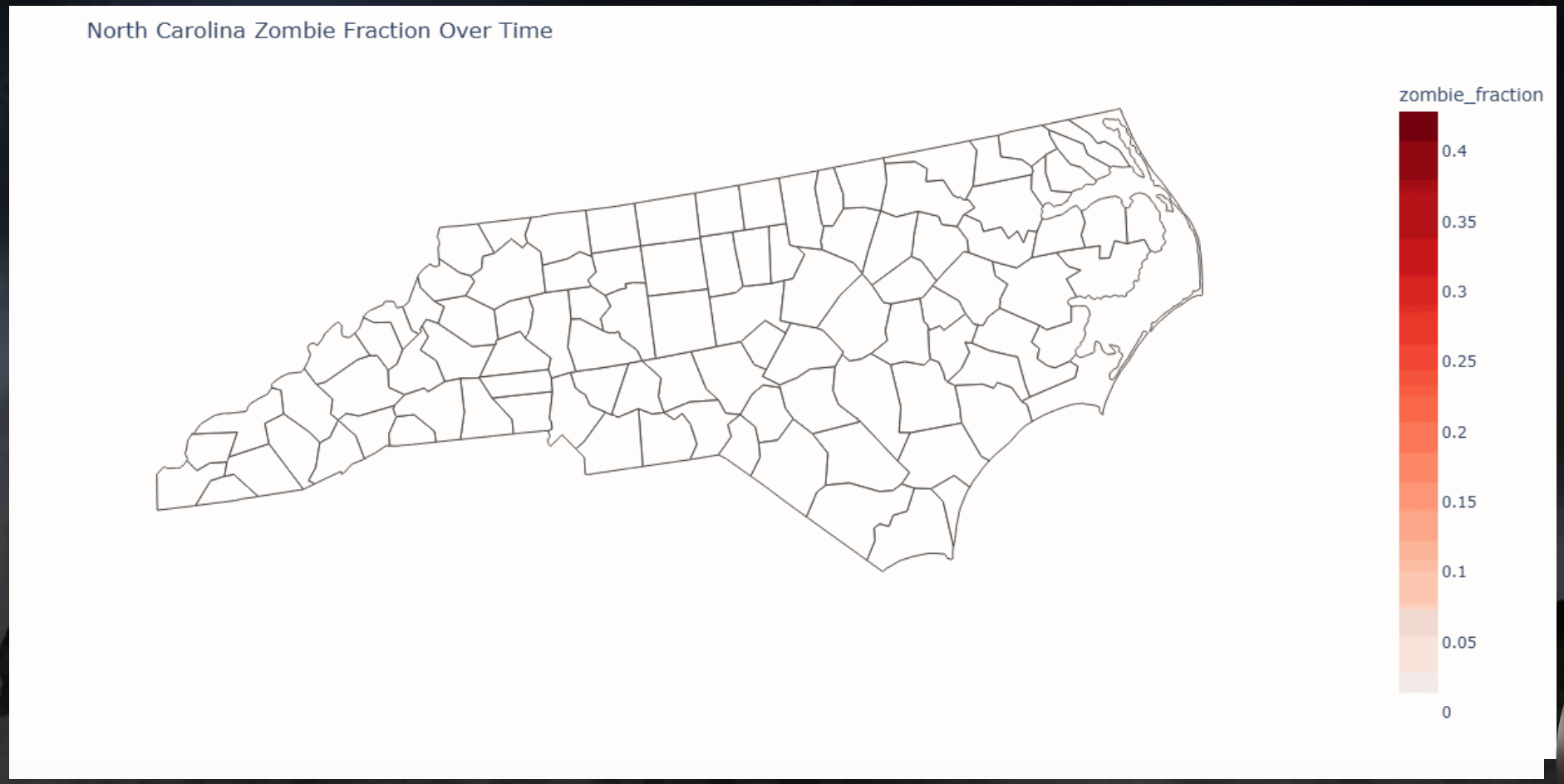


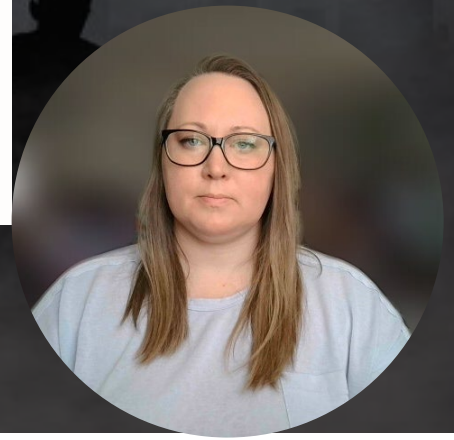


THE SPREAD


Statewide Zombie Spread (Every 30 Days, 3 Years)
Survival
Zombies
Average Fraction
Days
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
Populations Stabilized
(Threshold Reached)
Zombie Population
Takes Over
Significant Population
Impact

THE SPREAD





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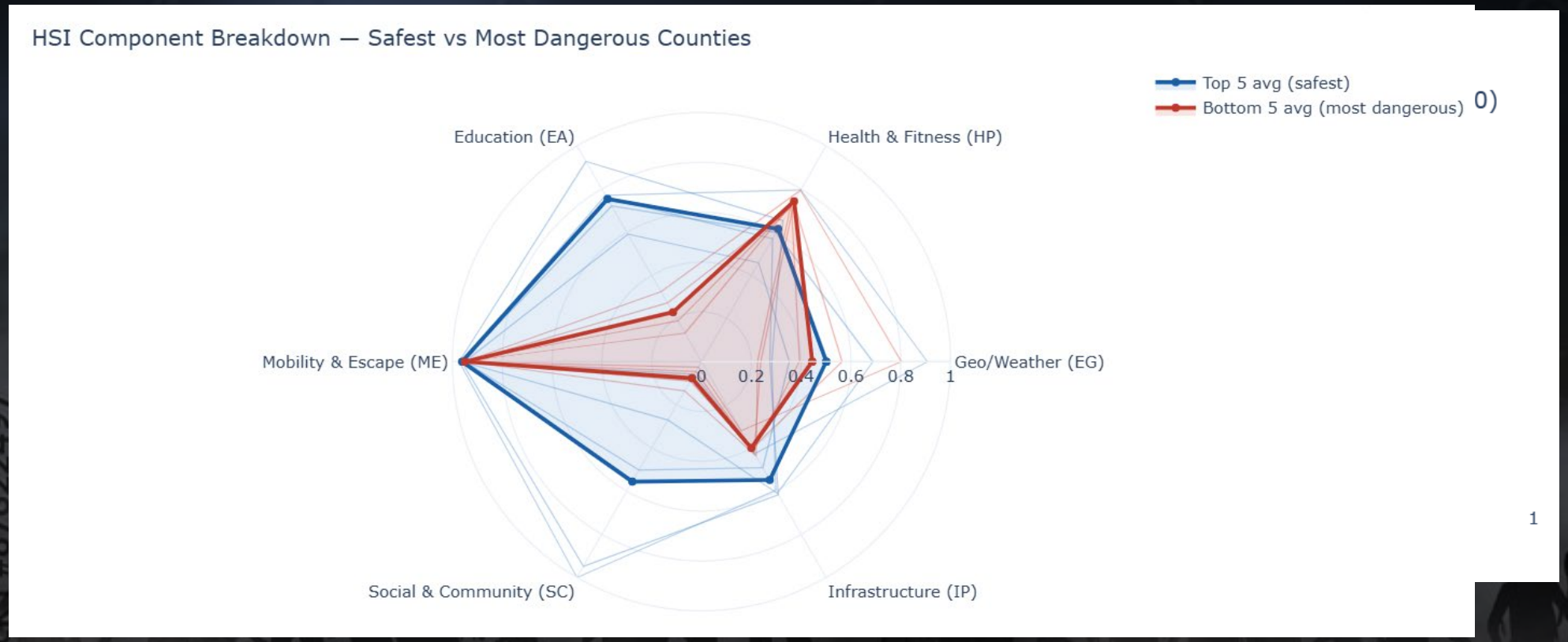
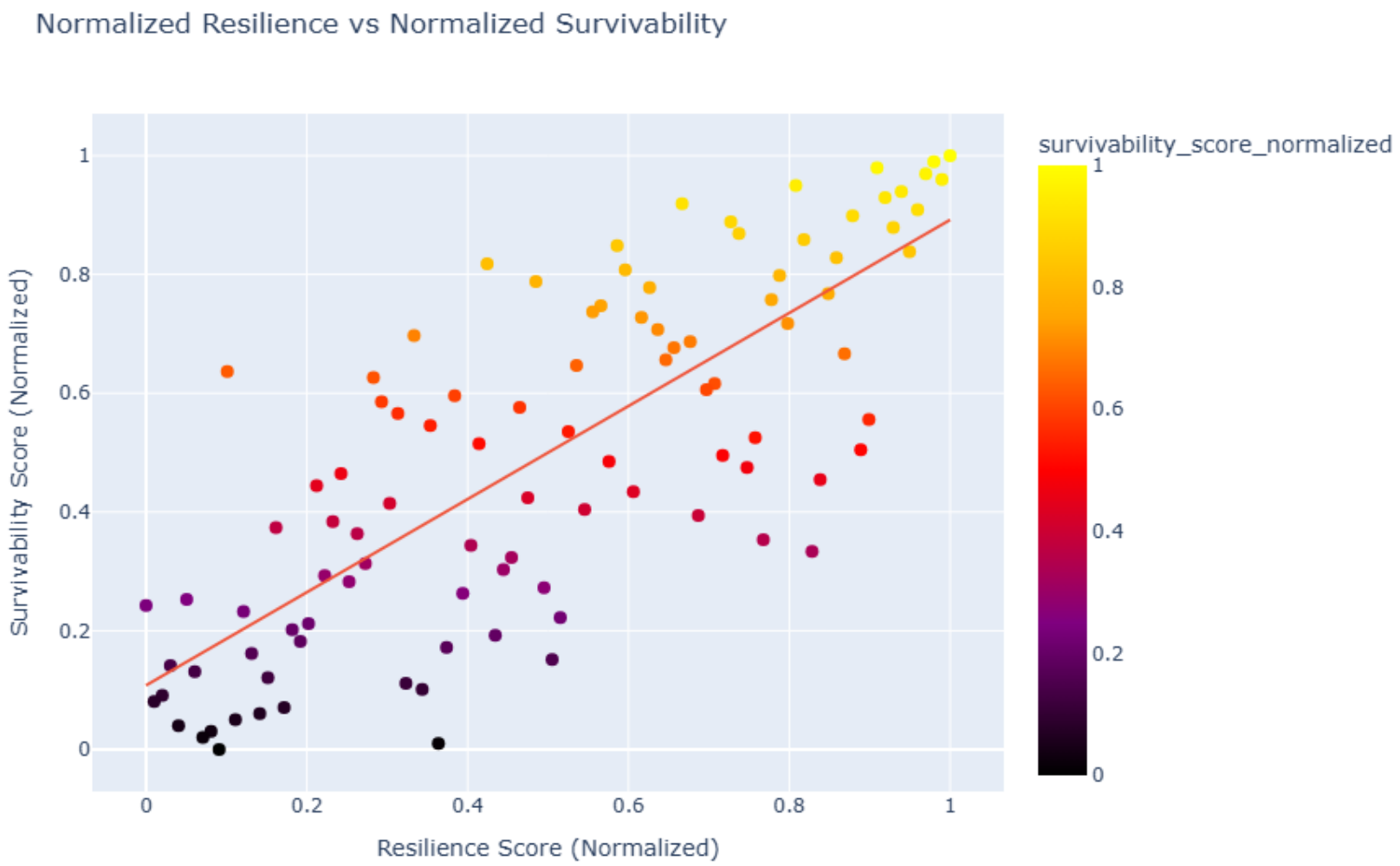
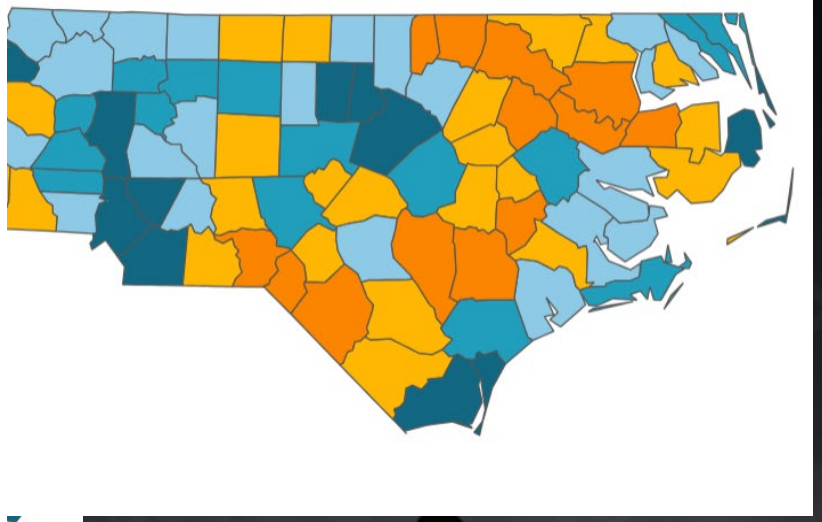
SURVIVAL INSIGHTS



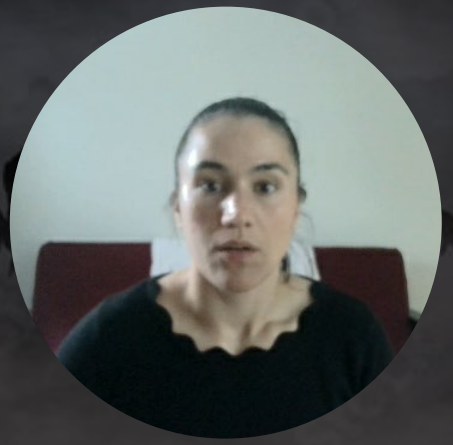
SURVIVAL INSIGHTS





Closing

- Fictional scenario, but a real epidemiological model based on standard S–I–R (Susceptible, Infected, Removed) framework
- Models help predict spread, evaluate risk, and guide response strategies
- Can adapt to different regions and conditions by adjusting parameters (β, κ)
- Even fictional outbreaks demonstrate real-world applications
- Modeling is essential for preparedness and decision-making
- Science drives understanding, and fiction helps us explore it



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